

Detection probability: a fixed reference scale and an estimated level

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Why this note exists

Supervisor meeting, 2026-09-25: ELVIS is approved for the counterfactual, but $\hat{\lambda}$ under linear detection, $q(e) = \lambda e$, is not credible (audit-triggering threshold $1/(2\hat{\lambda}) \approx \text{COP } 920,000$ against median reported materials of about 8,414). The diagnosis is a **scale problem**: one λ sets both the *level* of detection (the slope of q) and the *ceiling* $e < 1/(2\lambda)$, and cannot make detection plausible while leaving room for the tail of evaders. Heterogeneity across firms is not the issue; productivity ω and the cost shock ψ carry it.

Approach: start from theory and evidence, fix the scale outside the estimation, choose the simplest form consistent with both, and change one thing at a time. (An earlier version of this note proposed a Box–Cox family with an estimated shape; dropped for a theory-first specification. See “Alternatives considered”.)

Evidence

Details and sources in `Research-log/log.md`, 2026-09-25.

- **Detection rises with the amount evaded** (Ecuador’s SRI targeted clients by the amount and share of ghost deductions; Spain’s detection function is increasing in concealed income).
- **Detection rises with firm size relative to peers and saturates** (US audit coverage by asset class: below 1% up to \$10m in assets, rising to 60–67% at the top; Spain’s monitoring jumps at a size threshold).
- **Levels are low and bounded away from certainty** (Ecuador: at most ~10% of evaders whose supplier was identified were notified; only 25% of notified firms amended).

- **Colombia, 1974–1985: weak sanctions.** Perry and Cárdenas (1986) describe a rigid limitation period on the authority’s power to revise returns, two years under the regime they criticize (vol. 3, pp. 277–78); *silencio administrativo positivo*; an ineffective sanction system, “trivial” sanctions in some cases and inapplicably large ones in others; and recurrent amnesties (vol. 2, pp. 144–45). A caught firm repaid at most what it owed.

Reading: q is increasing and concave in the amount evaded relative to peers, $q(0) = 0$, and there is no effective fine beyond the tax owed.

The reference scale

$x \equiv e/\bar{M}_{j,t-1}$, where $\bar{M}_{j,t-1}$ is the mean (or median) **reported** materials M^* among all firms in industry j in year $t - 1$.

- **Lagged:** the firm can know it (last year’s reports, industry statistics, employees moving between firms), and it does not contain the firm’s own current M_{it} , so $\partial q/\partial M = 0$ still holds (the deconvolution needs it) and ELVIS’s latent M_{it} stays out of q . The lagged-observation sample already drops each firm’s first year.
- **Industry peers:** materials intensity differs across industries; authorities benchmark within sector.
- **Reported, not true, materials:** true M is unobserved for unincorporated firms and observed only for corporations, which are much larger; reported M^* is also what the authority compares against. As a predetermined constant it creates no endogeneity.
- **Units:** materials is nominal materials deflated by the GDP deflator (Code/Colombia/10_data_wrangling) the same units as e and M^* in `stage2_data`, so a lagged average carries no inflation gap.
- Common to all firms in an industry-year; not firm-specific.

Sensitivity: mean vs. median; industry vs. all manufacturing.

The detection function and the fine

$$q(e) = \lambda_1 (1 - e^{-x}), \quad x = \frac{e}{\bar{M}_{j,t-1}}, \quad 0 < \lambda_1 \leq 1.$$

$q(0) = 0$, increasing, concave, saturating at λ_1 . In this family the scale and the curvature are one parameter: $-q''/q' = 1/\bar{M}_{j,t-1}$. With the scale fixed, the curvature is fixed too, and the one estimated parameter is the **level** λ_1 , the maximum detection probability.

The fine is not separately identified. The evasion FOC is $\tau \rho_t [1 - (1 + \phi)(q + q'e)] = C(\omega, \psi)$ with $C \equiv \partial \kappa / \partial e$. With $q = \lambda_1 f(x)$, the fine and the level enter only as $\tilde{\lambda} \equiv (1 + \phi)\lambda_1$ (absorbed by the level, not by δ_0):

$$B(x) \equiv 1 - (1 + \phi)(q + q'e) = 1 - \tilde{\lambda} G(x), \quad G(x) = 1 - e^{-x}(1 - x).$$

Evidence pins ϕ : under the Colombian regime of the period a caught firm repaid at most the tax owed, so $\phi = 0$ and $\tilde{\lambda} = \lambda_1$. Weak enforcement (limitation periods, amnesties, slow processes) lowers the probability of being caught *and made to pay*: it belongs in q , not in ϕ .

SOC. $B'(x) = \lambda_1 e^{-x}(x - 2) < 0$ for $x < 2$: the local interior optimum is unique on $x < 2$.

Global optimality binds earlier, at $x < 1$. The firm's expected gain is $\pi(e) = e[\tau\rho_t(1 - \lambda_1) - C + \tau\rho_t\lambda_1 e^{-x}]$, and its FOC gives $C = \tau\rho_t B(x_i)$.

- If $x_i < 1$: $C = \tau\rho_t B(x_i) > \tau\rho_t B(1) = \tau\rho_t(1 - \lambda_1)$. For every $x > x_i$, $\pi'(e) = \tau\rho_t B(x) - C < 0$ (B falls until $x = 2$, then rises back only towards $1 - \lambda_1 < C/\tau\rho_t$). So x_i is the **global** optimum, for any $\lambda_1 \leq 1$.
- If $x_i > 1$: $C < \tau\rho_t(1 - \lambda_1)$, and the firm gains from evading without limit.

No firm evades without limit in the data, so the model implies every evader has $e < \bar{M}_{j,t-1}$: a restriction on the support of the latent variable (equivalently, costs bounded below, $C \geq \tau\rho_t(1 - \lambda_1)$, a restriction on the distribution of ψ , which ELVIS leaves free), not on the parameter. With $\lambda_1 = 1$ the restriction holds automatically ($B(1) = 0$), so fixing $\lambda_1 = 1$ is not needed.

Implications: among optimizing evaders, detection is at most $q(\bar{M}_{j,t-1}) = (1 - e^{-1})\lambda_1 \approx 0.63\lambda_1$, reached by a firm whose overreported materials equal the average firm's materials bill; $B > 0$ on the whole support for $\lambda_1 < 1$.

Specification:

$$h(e) = \ln \tau_{it} + \ln \rho_t + \ln[1 - \lambda_1 G(x)], \quad x = \frac{e}{\bar{M}_{j,t-1}} \in [0, 1).$$

The external evidence (at most ~10% of evaders notified in Ecuador; 60–67% audit coverage at the top of the US size distribution) is a check on $\hat{\lambda}_1$, not an input.

What changes in ELVIS

- **Parameters:** $\theta = (\lambda_1, \delta_0, \delta_1, \delta_2)$: same count as today, but the detection parameter is the level, not the scale.
- **Sampler support:** $M \in (\max\{0, M_{it}^* - \bar{M}_{j,t-1}\}, M_{it}^*)$, fixed per industry-year and independent of λ_1 . The support no longer moves with a parameter, which removes the mechanism behind the thin-tail identification of λ and the η episode; λ_1 is profiled smoothly like the δ 's, with no grid.
- **Moments:** rows 1–8 unchanged in form (with the new h); row 9, the score moment for the detection parameter, becomes

$$\frac{\partial h}{\partial \lambda_1} \varepsilon = -\frac{G(x)}{1 - \lambda_1 G(x)} \varepsilon,$$

bounded for $\lambda_1 < 1$ because $G \leq 1$ on $x < 1$ (denominator $\geq 1 - \lambda_1$); no transform needed. $d_g = 9$.

- **Identification of λ_1 :** how the marginal benefit B declines with evasion relative to the reference scale, across all firms, rather than from a few firms near a common ceiling in levels.

Year effects for ρ_t (step 2, separately)

$\ln(\tau\rho_t) = \ln\tau_{it} + \ln\rho_t$ (Salvador), and ρ_t , the price of materials relative to the GDP deflator, is not observed. Since $\psi = h(e) - \delta_0 + \delta_1\omega - \delta_2\omega^2$, a year-specific $\ln\rho_t$ is absorbed by year-specific intercepts:

$$\psi = \ln\tau_{it} + \ln[1 - \lambda_1 G(x)] - \delta_{0,t} + \delta_1\omega - \delta_2\omega^2, \quad \delta_{0,t} \equiv \delta_0 - \ln\rho_t.$$

- **Moments:** replace $E[\psi] = 0$ with $E[\psi \cdot 1\{\text{year} = t\}] = 0$ for each of the 11 years (1981–91). The other ψ rows stay as they are: ψ is mean zero within each year, so its products with ω , ω^2 , $\ln M$ remain valid (the anchor rule in CLAUDE.md). - **Count:** θ gains 10 parameters (11 intercepts replace one), d_g gains 10 rows; the γ 's do not count against the order condition. - **Identification of λ_1** then comes from variation in τ_{it} across industries and firms within a year. - **What stays in ψ :** industry-by-year variation in the relative price of materials. If it matters, industry-by-year intercepts would absorb it, at many more parameters; not the first step. - **In the counterfactual:** ρ_t cancels (the FOC is taken relative to the baseline), so the year effects change the estimates, not the counterfactual formula.

What changes in the counterfactual

With $\tilde{\tau} = (1 + \Delta)\tau$ and the baseline FOC $C = \tau\rho_t B(x_i)$, the counterfactual FOC is $B(x') = B(x_i)/(1 + \Delta)$. B is strictly decreasing on $[0, 1)$, from 1 to $1 - \lambda_1$, so

$$e'_i(\Delta) = \begin{cases} 0, & B(x_i)/(1 + \Delta) \geq 1, \\ \bar{M}_{j,t-1} B^{-1}(B(x_i)/(1 + \Delta)), & 1 - \lambda_1 < B(x_i)/(1 + \Delta) < 1, \end{cases}$$

solved by bisection on $[0, 1)$.

- **Open issue: a tax increase can push firms past $x = 1$.** If $B(x_i)/(1 + \Delta) \leq 1 - \lambda_1$, the firm's new optimum is unbounded: the higher rate lowers its effective cost below $\tau\rho_t(1 - \lambda_1)$. The affected firms are those already near the ceiling. For $\Delta = +0.5\%$, only firms with x_i above about 0.89 ($\lambda_1 = 0.1$), 0.99 ($\lambda_1 = 0.5$) or 0.998 ($\lambda_1 = 0.9$) are affected; for $+2\%$, above 0.65, 0.95 and 0.994. With $\lambda_1 = 1$ there is no such firm. Report the share of firms affected at each tested Δ once $\hat{\lambda}_1$ is known; if it is not negligible, it is a finding in itself (at the margin, weak enforcement lets high-rate evasion explode) to discuss with supervisors, not something to fix by tuning.

- The uniform-shift property of linear q is gone: the response depends on the firm's own x_i , so the cut/increase asymmetry has to be re-derived, not assumed.
- Revenue: claimed deductions $\tilde{\tau}[M_i + (1 - q(e'_i))e'_i]$ as before.

Alternative: power form with a fixed scale

$q(e) = \min\{x^a, 1\}$, $x = e/\bar{M}_{j,t-1}$, $0 < a \leq 1$ (here the estimated parameter a is the *shape*; the level is tied to the scale, $q = 1$ at $e = \bar{M}_{j,t-1}$).

- **Nests the current linear q at $a = 1$** , now with the scale fixed: a clean bridge from today's model.
- SOC holds everywhere: $-q''e/q' = 1 - a < 2$.
- Well-posed automatically: q reaches 1 at $x = 1$, and beyond it evading only costs.
- **But the ceiling moves with a** : $B = 1 - (1 + a)x^a > 0$ requires $x < (1 + a)^{-1/a}$, between 0.37 ($a \rightarrow 0$) and 0.5 ($a = 1$), so the sampler support depends on the parameter again (in units of $\bar{M}$); detection at the ceiling is $1/(1 + a) \in [0.5, 1)$, high relative to the evidence; and marginal detection is infinite at $e = 0$ for $a < 1$.

Kept as the alternative; the exponential is the step-1 specification because its support does not move with the parameter and its level is free.

Order of changes

0. Reproduce the current headline (linear q , single tax, current PF).
1. New q only (exponential, $\bar{M}$ as lagged industry mean; median as sensitivity).
2. Add year effects.
3. Two-tax model, net-of-tax first stage, joint efficient-GMM PF.

Checks before code

1. Build $\bar{M}_{j,t-1}$ from `stage2_data` (`M_star` by `sic_3`, lagged one year); report it by industry.
2. Distribution of $M_{it}^*/\bar{M}_{j,t-1}$ for unincorporated firms: how many can have e near the ceiling.
3. Verify numerically: B , B' , the global-optimality argument, the score's bound, and the counterfactual bisection on test values (done for the formulas above).
4. Then: sampler support and moments in `Code/Rcpp/1200-stage2-elvis-common.h` / `Code/C-estimator/grid_estimator.cpp`.

Alternatives considered

- **Box–Cox family** $q = \lambda_1[(1+x)^k - 1]/k$ (nests linear, log, bounded concave) and **Weibull** $q = \lambda_1(1 - e^{-x^k})$ (S-shaped for $k > 1$): flexible, but they add a parameter chosen by fit rather than by theory.
- **Firm-specific scale** $\lambda_{0,i}$: harder to justify, and risks reintroducing the firm's own M into q .